A Label Placement

While VERA exracts semantically meaningful visual explanations, there remains the practical apsect of how to visualize these visual explanations on-screen. A key challenge in this is how to position region labels, so that each region polygon is accompanied by a textual description of what the region represents. Label placement has been a well-studies problem, with several approaches being proposed over the years. However, these algorithms typically deal with annotating specific points on a scatter plot and are not directly applicable to our task here. However, extending any one of these is straightforward to our particular task.

Label Placement Algorithm. We here describe a force-directed label placement algorithm we designed for VERA. A good label positioning for a VERA explanation must necessarily fulfill the following conditions:

- Labels must be inside the axes bounding box and must not overflow.
- Labels must not overlap one another as overlapping text is often completely illegible.
- Labels should not overlap with the regions, since labels may be illegible.
- Labels should be positioned closely to their associated region.

Drawbacks and Limitations. We here describe the algorithm.